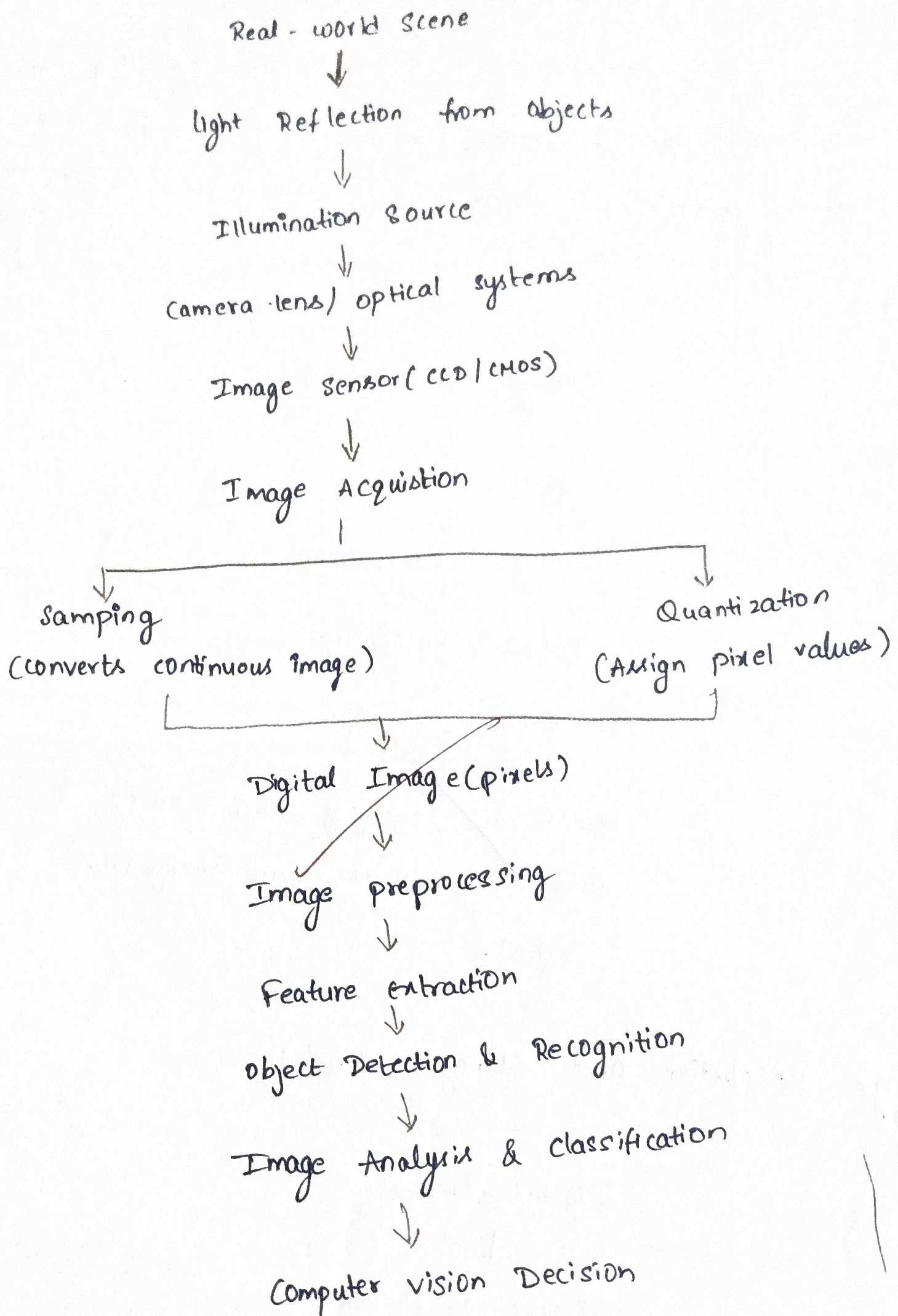


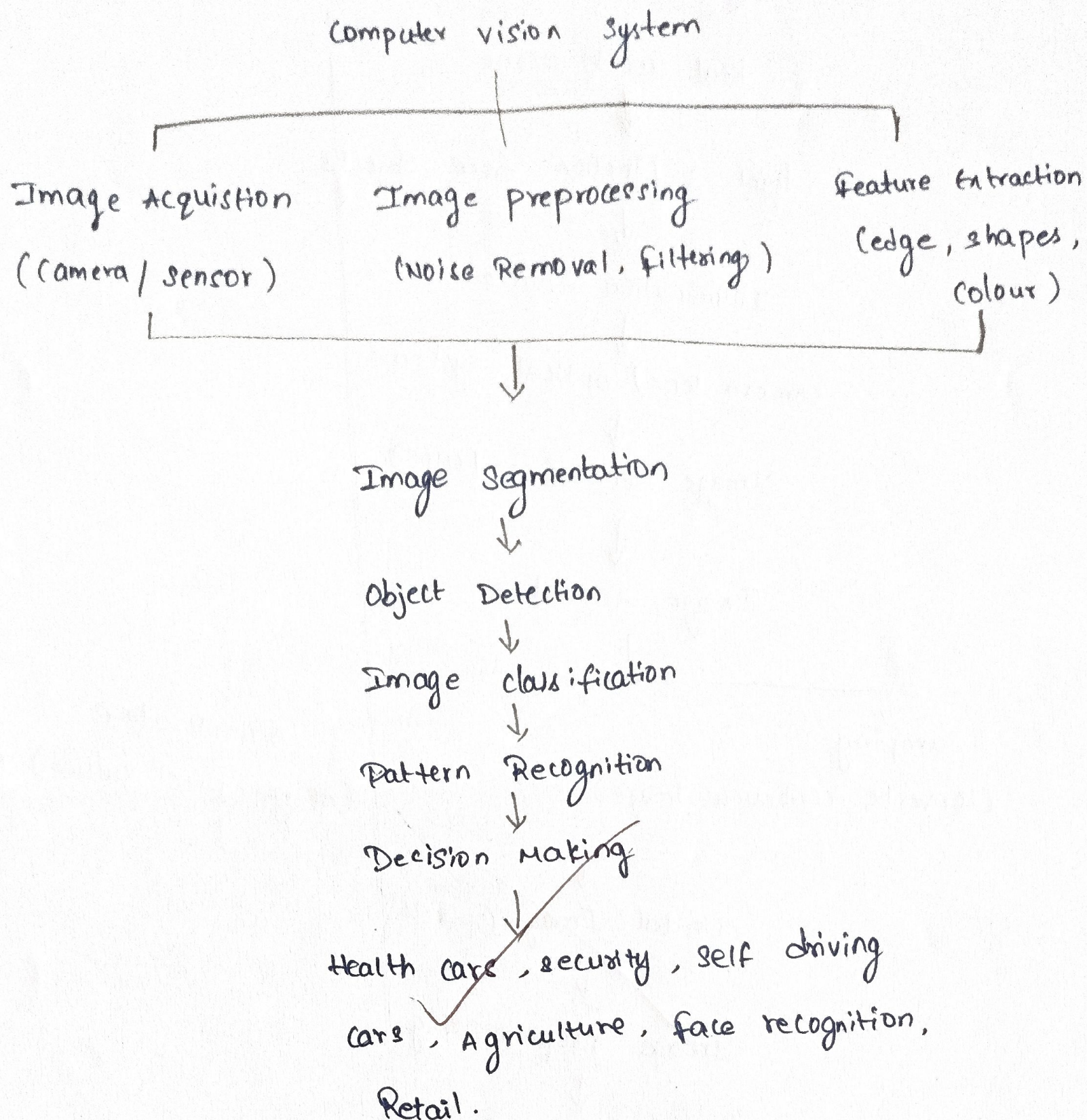
① Real-world scene to Digital Image presentation





2)

## Computer vision system pipeline.



3)

## Image Resolution and Image Quality

Digital Image



Image Resolution





Spatial Resolution  
(Number of Pixels)  
Higher Pixels → Better Detail

Intensity Resolution  
(Brightness Levels)  
More Levels → Better Contrast

Image Quality

Better Accuracy ↓ clear objects ~~more details~~

Applications

- Medical Imaging
- Satellite Images
- Face Recognition
- Traffic Monitoring
- Autonomous Vehicles

4) Factors Affecting Image Acquisition

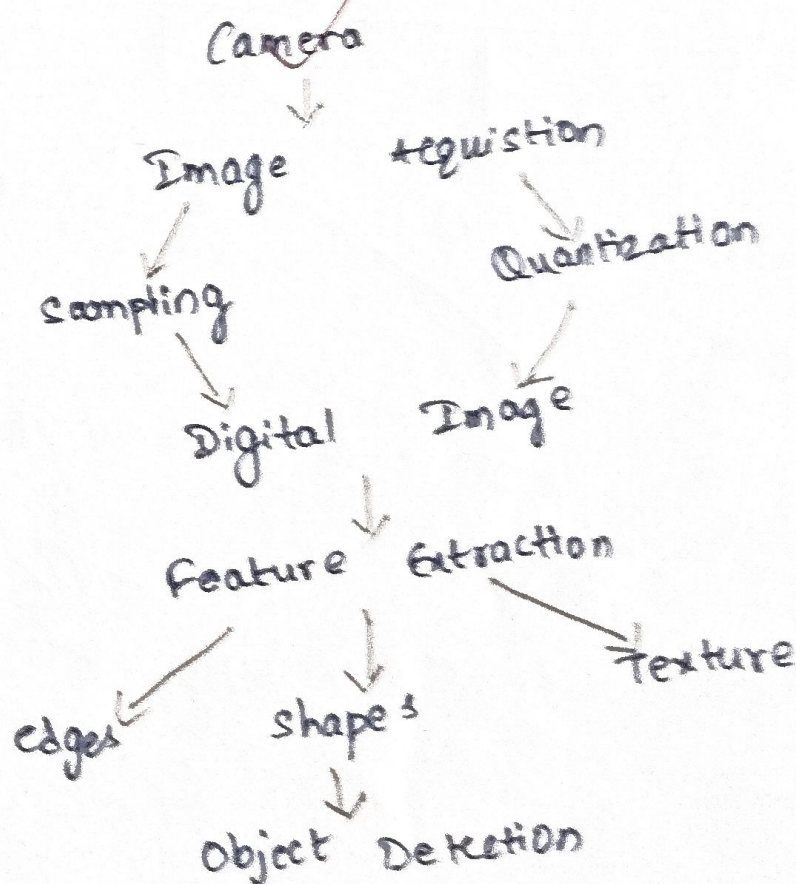




Image classification

↓  
Computer vision

↓  
Real Application

Computer vision Capabilities

COMPUTER VISION SYSTEM

↓  
Image Acquisition

↓  
Image processing

↓  
Feature processing

↓  
Feature extraction

↓  
Computer vision capabilities

object Detection

face  
Recognition

Image  
classification

↓  
Healthcare

↓  
surveillance

↓  
vehicles

↓  
Agriculture

↓  
Retail

↓  
Better Accuracy & Automation

↓  
Faster decision making